Lesson 3: Random Functions and Dice Games

Module 1 | 60 minutes | random, simulation, turn functions | Grades 7–11 · ages 12–16

Name: _____ Class: _____ Date: _____

Practise what you learned today. Read each task carefully.

1. Key words

Match each word to what it means. Write the letter.

a. random.randint(a, b) ___ A. A loop that keeps the game going until someone wins.
b. simulation ___ B. How likely something is, like 1 in 6 for rolling a six.
c. probability ___ C. Using a program to try something many times and see what happens.
d. game loop ___ D. Gives a random whole number from a to b, including both.

2. Plan your program

Before you type it: Dice simulation. Fix roll_dice(sides) so it returns random.randint(1, sides) (using the parameter!). Then roll a six-sided dice 100 times with a loop, count the sixes in a variable called sixes, and print `Sixes: ... out of 100`. Write the steps in plain English (pseudocode), then the Python you will use.

3. Predict the output: two dice

Which of these outputs is IMPOSSIBLE? Code: import random ⏎ ⏎ def roll(): ⏎ return random.randint(1, 6) ⏎ ⏎ print(roll() + roll())

a. A. 1
b. B. 2
c. C. 7
d. D. 12

4. Will this give the same answer every time you run it?

Write each item in the right group: Same every run / Can change every run.

a. 3 + 4 → _____
b. len("ludo") → _____
c. max(2, 9) → _____
d. random.randint(1, 6) → _____
e. random.choice(["heads", "tails"]) → _____
f. roll_dice(6) when it uses randint → _____

5. Plan your program

Before you type it: Debug it! This program should print `You rolled ...` and then `Double it: ...` with a real number. It crashes. Find and fix the two bugs. Write the steps in plain English (pseudocode), then the Python you will use.

How did it go? Colour a face: 😀 I can do it · 🙂 Nearly there · 🤔 I need help

☐ I can write a function that returns a random result.
☐ I can use a loop to simulate many rolls and count results.
☐ I can write a rule function and test it with fixed numbers.
☐ I can split a game turn into its own function.